

Module 4: Supervised Learning
Submodule: 4.4 Generalized Linear Models
Lecture: Exponential Family Models

XCS229 Machine Learning

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I. Introduction to Exponential Family Models

Exponential family models are an important topic in our lecture today. These models are particularly interesting because they provide a powerful framework for understanding various machine learning models such as regression and classification.

This unification allows us to apply similar techniques across different models repeatedly, enhancing our understanding and making our learning process more efficient.

The main goal of this lecture is to unify inference and learning processes for a variety of models. By decoupling the model's components, we can understand how to derive insights and make predictions effectively.

II. Definitions

The exponential family consists of a specific set of probability distributions. Its significance lies in its broad applicability across various fields. A probability distribution $P(y; \eta)$ is defined in the **exponential family** as follows:

$$p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta)).$$

Where:

- y : the data.
- η : are the natural parameters (also called the canonical parameters) of the distribution.
- $T(y)$: is the sufficient statistic, which condenses all the relevant information about the data y into a simplified form. (For the distributions we consider, it will often be the case that $T(y) = y$).
- $b(y)$: is a base measure that outputs a scalar and only depends on the data and does *not* depend on the natural parameters η .
- $a(\eta)$: is a log partition function that outputs a scalar and does *not* depend on the data y .

The quantity $e^{-a(\eta)}$ essentially plays the role of a normalization constant, making sure the distribution $p(y; \eta)$ sums/integrates over y to 1.

It is important to note that not every distribution fits into this framework, but many common distributions do.

The interactions between these components are significant. For instance, $T(y)$ and η must have the same dimension since we're taking the dot product of the two. This means that the dimensionality of the sufficient statistics must align with the natural parameters for valid operations.

III. Examples

Bernoulli Distribution Example

The **Bernoulli distribution** is defined as:

$$P(y; \phi) \triangleq \phi^y (1 - \phi)^{1-y}$$

To express the function in an exponential form, we take the logarithm:

$$\begin{aligned} p(y; \phi) &= \phi^y (1 - \phi)^{1-y} \\ &= \exp(y \log \phi + (1 - y) \log(1 - \phi)) \\ &= \exp\left(\left(\log\left(\frac{\phi}{1 - \phi}\right)\right) y + \log(1 - \phi)\right) \end{aligned}$$

Now, let's identify each component of the Bernoulli distribution in the context of the exponential family. Identifying the components:

- $T(y) = y$
- $\eta = \log\left(\frac{\phi}{1-\phi}\right)$
- $b(y) = 1$
- $a(\eta) = -\log(1 - \phi)$

Claim: $a(\eta) = -\log(1 - \phi)$.

Proof. We start with the expression for the natural parameter η :

$$\eta = \log\left(\frac{\phi}{1-\phi}\right)$$

We can rewrite ϕ as follows:

$$\phi = \frac{1}{1 + e^{-\eta}}$$

Now, we calculate $1 - \phi$:

$$1 - \phi = 1 - \frac{1}{1 + e^{-\eta}} = \frac{e^{-\eta}}{1 + e^{-\eta}}$$

Rearranging gives us:

$$1 - \phi = \frac{1}{1 + e^{\eta}}$$

From the expression $1 - \phi = \frac{1}{1+e^{\eta}}$, we take the logarithm:

$$-\log(1 - \phi) = -\log\left(\frac{1}{1 + e^{\eta}}\right)$$

Applying logarithm rules yields:

$$-\log(1 - \phi) = \log(1 + e^{\eta})$$

Thus, we have shown that:

$$a(\eta) = -\log(1 - \phi) = \log(1 + e^{\eta}).$$

□

Gaussian Distribution Example

Recall that, when deriving linear regression, the value of σ^2 had no effect on our final choice of θ and $h_\theta(x)$. Thus, we can choose an arbitrary value for σ^2 without changing anything. To simplify the derivation below, let's set $\sigma^2 = 1$.

Thus, the **Gaussian distribution**, assuming a fixed variance ($\sigma^2 = 1$), is given by:

$$p(y; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y - \mu)^2\right)$$

We can rewrite this in the exponential family form:

$$\begin{aligned} p(y; \mu) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}y^2 + \mu y - \frac{1}{2}\mu^2\right) \\ &= \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}y^2) \cdot \exp\left(\mu y - \frac{1}{2}\mu^2\right) \end{aligned}$$

Thus, we see that the Gaussian is in the exponential family, with the components:

$$\begin{aligned} \eta &= \mu \\ T(y) &= y \\ a(\eta) &= \frac{\mu^2}{2} = \frac{\eta^2}{2} \\ b(y) &= \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}y^2) \end{aligned}$$

IV. Why Exponential Family Models Matter

One significant advantage of the exponential family is the simplification it brings to inference:

$$\mathbb{E}[y, \eta] = \frac{\partial a(\eta)}{\partial \eta}$$

This equation shows that the expectation can be computed by the derivative of the log partition function.

The maximum likelihood estimation (MLE) procedure becomes more tractable because the function representing the likelihood is concave. This means that minimizing the negative log likelihood leads to a convex optimization problem, allowing for easier calculation of parameter estimates.

Convex functions have a unique property where every local minimum is also a global minimum, ensuring that we can confidently find optimal model parameters.

V. Summary

In this lecture, we covered essential aspects of exponential family models, highlighting their importance in machine learning applications. Key take-aways include:

- **Definition of Exponential Family:** We defined the exponential family of distributions, emphasizing their structure and components: the natural parameter η , the sufficient statistic $T(y)$, the base measure $b(y)$, and the log partition function $a(\eta)$.
- **Examples:**
 - The **Bernoulli Distribution** was analyzed, where we identified its components and demonstrated a proof for the function $a(\eta) = -\log(1 - \phi)$.
 - The **Gaussian Distribution** was examined, illustrating how it fits within the exponential family framework with fixed variance and specifying its parameters.
- **Importance of Inference:** We highlighted the simplification of inference through the relationship between the expectation and the derivative of the log partition function, leading to convenience in maximum likelihood estimation (MLE) procedures.
- **Convexity in Learning:** The significance of convex optimization in parameter estimation was discussed, ensuring unique solutions for optimal parameters, which simplifies the learning process.